

San Aung

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Software Engineering student at King Mongkut's Institute of Technology Ladkrabang with hands-on experience in **data processing, event-driven pipelines, NLP, AI systems, and backend engineering**. Experienced with Python, SQL, Docker, cloud deployment, and production-oriented backend systems.

Education

- **Bachelor of Engineering in Software Engineering, KMITL (GPA 3.4 / 4.0) 2024 - Present**
- **Bachelor of Information And Communication Technology, UTYCC (Myanmar) 2017- 2020**

Experience

Software Engineering Intern - ATI Limited - Bangkok

April 2026 – June 2026

- Automated **data cleaning, transformation, EDA, and visualization workflows** using Python, pandas, and NumPy, reducing repetitive manual data preparation and making datasets ready for downstream analysis.
- Built automated **web scraping pipelines** to collect and structure external data, enabling consistent downstream analysis and reporting without manual data collection.
- Developed a **newsletter subscription platform** with reusable backend APIs, subscription management, administrative controls, and automated email workflows, providing a centralized system for managing subscriber operations.
- Deployed a **containerized AI assistant** on a Linux VPS using Docker Compose, Nginx, DNS, and HTTPS, turning the assistant into an accessible production service.

Computer Architecture and Organization Teaching Assistant – King Mongkut's Institute of Technology Ladkrabang

July 2026 – Present

- Assisted the Professor in laboratory sessions for approximately **50 second-year students**.
- Provided debugging support and code review guidance for **ARM assembly and C programming** lab exercises.
- Assisted students with **Raspberry Pi** setup, development environment configuration, and managing lab activities.

Projects

Bangkok Flood Risk Map | [Kotlin](#), [Spring Boot](#), [jOOQ](#), [PostgreSQL/PostGIS](#), [Next.js](#), [TypeScript](#), [Material UI](#), [MapLibre](#), [Gemini API](#), [Docker Compose](#) (Source code: [GitHub](#))

- **Geospatial Engine & PostGIS Architecture:** Built a full-stack flood-risk mapping application processing **~19,900 OpenStreetMap** building geometries across **3 Bangkok districts**, combining PostGIS spatial queries, river proximity, and building heights to estimate flood exposure.
- **High-Performance API & Query Optimization:** Designed a Kotlin + Spring Boot REST API using jOOQ over PostGIS to serve GeoJSON payloads; optimized query performance from 6–14s down to **<1s (over 90% reduction)** by implementing spatial indexes (**GIST**) and materialized views.
- **AI-Driven Spatial Querying & Safety:** Integrated **Gemini API** to convert natural language queries into validated spatial filter parameters executed via parameterized SQL, incorporating

keyword fallbacks, response caching, and per-IP rate limiting to prevent SQL injection and LLM hallucinations.

- **Frontend & One-Command Deployment:** Developed a Next.js and Material UI frontend featuring interactive MapLibre risk heatmaps and building-level popups; containerized the complete application stack with Docker Compose for single-command orchestration.

Kafka News Intelligence Pipeline | [Apache Kafka](#), [FastAPI](#), [PostgreSQL](#), [Qdrant](#), [Docker](#), [HuggingFace Transformers](#), [Chart.js](#) (Source code: [GitHub](#))

- Built an **event-driven news processing pipeline** with Apache Kafka, enabling asynchronous ingestion and processing of continuously arriving news data.
- Designed independent services for **ingestion, persistence, sentiment analysis, vector embeddings, and analytics**, improving separation of responsibilities and allowing components to operate independently.
- Integrated **RoBERTa** for automated sentiment classification and **Qdrant** for vector storage and semantic search, transforming raw news into searchable and analyzable information.
- Developed FastAPI services and a Chart.js dashboard to make processed news intelligence accessible through APIs and visual analytics.
- Containerized the complete system with **Docker Compose** and deployed Kafka and **PostgreSQL** using managed cloud infrastructure, making the multi-service system reproducible across environments.

Solvehub | Used [React.js](#), [Tailwind CSS](#), [FastAPI](#), [MySQL](#), [JWT](#), [Bcrypt](#), [Axios](#), [SQLAlchemy](#) (Source code: [GitHub](#))

- Developed a community-driven Q&A platform that centralizes technical questions and knowledge sharing in a structured, searchable application.
- Built **REST APIs** with **FastAPI** and **SQLAlchemy** and implemented **JWT authentication** and MySQL persistence to support secure user interactions and data management.

Portfolio | Used [React.js](#), [Next.js](#), [TypeScript](#), [Vercel hosting](#) (Source code: [GitHub](#))

- A high-performance, fully responsive portfolio website that highlights my web development journey and key projects

ISP Myanmar Sentiment Analysis | Used [pandas](#), [BeautifulSoup](#), [nltk](#), [VADER](#), [matplotlib](#), [seaborn](#), [spaCy](#)

- Built an automated pipeline to **scrape, clean, and structure event data** from the ISP Myanmar website, replacing manual collection with a repeatable workflow.
- Applied NLP-based **sentiment analysis, speaker bias analysis, and keyword analysis** to transform unstructured event information into structured insights.
- Created visualizations and analytical reports that surfaced **sentiment trends and potentially significant events**, making patterns easier to identify and interpret.

Skills

- **Programming Languages:** JavaScript, TypeScript, Python, Java, Kotlin, SQL
- **Frontend:** Material UI, HTML, CSS, Bootstrap, React.js, Next.js, Tailwind CSS
- **Backend:** Spring Boot, JOOQ, Node.js, Express.js, FastAPI, REST APIs, Drizzle ORM
- **Databases:** MySQL, PostgreSQL(PostGIS), MongoDB, Redis, Supabase, Qdrant

- **Data Analytics & Visualization Tools:** pandas, NumPy, matplotlib, seaborn, Google Sheets
 - **Web Scraping:** BeautifulSoup, Requests
 - **Machine Learning & NLP:** Gemini API, scikit-learn, spaCy, VADER, nltk
 - **Version Control & CI/CD:** Git, GitHub Actions
 - **Infrastructure & Deployment:** Docker, Docker Compose, Apache Kafka, Linux VPS, Nginx, Railway, Netlify, Vercel
 - **Testing:** Pytest, Unit Testing
 - **Tools:** GitHub Copilot, Claude Code, OpenCode
 - **Others :** Object Oriented Principles(OOP), Data Structures and Algorithms
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Certifications

- [CS50's Introduction to Computer Science](#)
- [Meta Full-Stack Developer Specialization](#)
- [Data Analyst in Python](#)
- [Associate Data Scientist](#)